

Rational quadrature: a denominator-depth refinement

Companion to the general-prime lower-bound note

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Use the notation and the classical unit-indicator polynomial $H_{r,p}$ from `../rational_designs/p_adic_lower_bound.tex`. The following refinement uses exactly the same composition argument.

Lemma 1. *For a positive integer N divisible by p^a , $a \geq 1$, put*

$$E_{r,p,a} = \left\lceil \frac{(1 - p^{-a})(r + 1)}{p - 1} \right\rceil - 1.$$

Then

$$v_p \left(\int_0^1 H_{r,p}(Nx) dx \right) \geq E_{r,p,a}.$$

Proof. For a primitive p th root of unity ζ , let $\pi_h = \zeta^h - 1$ and normalize $w(\pi_h) = 1$, so $w(q) = (p - 1)v_p(q)$ for rational q . Set $\alpha = 1 - p^{-a}$. Every positive-degree coefficient of

$$W_{N,h}(t) = (1 + \pi_h t)^N - 1$$

satisfies

$$w([t^k]W_{N,h}) \geq \lceil \alpha k \rceil + 1.$$

For $k \geq p^a$ the factor π_h^k is enough, since $\alpha k \leq k - 1$. For $1 \leq k < p^a$, the identity $\binom{N}{k} = (N/k)\binom{N-1}{k-1}$ shows that $p \mid \binom{N}{k}$; thus the valuation is at least $k + p - 1 \geq \lceil \alpha k \rceil + 1$.

Write $G(z) = z/\log(1 + z) = \sum_j g_j z^j$. The companion note proves $w(g_j) \geq -j$. Consequently the coefficient $A_{k,h}$ in

$$\int_0^1 (1 + \pi_h t)^{Nx} dx = G(W_{N,h}(t)) = \sum_k A_{k,h} t^k$$

has valuation at least $\lceil \alpha k \rceil$. The same uniform-convergence argument used in that note gives $\sum_k A_{k,h} = 1$ at $t = 1$, because $p \mid N$. Therefore

$$\int_0^1 H_{r,p}(Nx) dx = \frac{1}{p} \sum_{h=1}^{p-1} \sum_{k>r} A_{k,h}$$

has w -valuation at least $\lceil \alpha(r+1) \rceil - (p-1)$. It is rational, so its integral-valued p -adic valuation is at least $\lceil \alpha(r+1)/(p-1) \rceil - 1$, as claimed. $\square$

Theorem 2. *Suppose rational nodes $x_1, \dots, x_K$ give a degree- r equal-weight uniform interval rule. Fix a prime p , and suppose*

$$a = \max_i \max(0, -v_p(x_i)) \geq 1.$$

Then

$$K \geq p^{E_{r,p,a}}.$$

More precisely, if q is the number of indices i with $v_p(x_i) = -a$, then

$$p^{\min(\lfloor r/(p-1) \rfloor, v_p(K) + E_{r,p,a})} \mid q.$$

Proof. The scaled nodes $y_i = p^a x_i$ are p -integral, and at least one is a unit. Apply quadrature exactness to $H_{r,p}(p^a x)$ and the unit filter modulo $p^{\lfloor r/(p-1) \rfloor}$. The lemma gives the divisibility. Since $E_{r,p,a} \leq \lfloor r/(p-1) \rfloor$ and $1 \leq q \leq K$, the displayed lower bound follows. $\square$

For $a = 1$, this recovers $E_{r,p,1} = \lfloor r/p \rfloor$. For $p = 2, a = 2$, the exponent is $\lceil 3(r+1)/4 \rceil - 1$, asymptotic to $3r/4$. For increasing denominator depth it approaches $r/(p-1)$. The statement is conditional on the actual largest denominator valuation; it does not improve the unconditional exponential lower bound because that valuation may equal one.